








Blockchain Applications

Lecture 7: Machine Learning, AI and Their Limits; Exam Review



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Today's Question and Route

- **Question**: machine learning (ML) and pretrained artificial intelligence (AI) models are now standard tools. Do they forecast crypto returns and crypto risk better than the statistical models of Lectures 2–6, and how do we find out without fooling ourselves?
- **Route**
  - can ML predict the direction of Bitcoin? Logit, random forest, gradient boosting
  - data snooping: what happens when we try 810 trading rules
  - risk forecasts: classical, ML and pretrained (foundation) models for Value-at-Risk
  - the limits of ML and AI in crypto finance
  - exam review: what you should be able to interpret
- Data: EOD Historical Data (EODHD) daily Bitcoin prices and volumes, 2015 – 23 October 2026

Learning Outcomes

- Set up a walk-forward evaluation and explain why in-sample accuracy is not evidence
- Read accuracy, the Brier score, the area under the curve and the Pesaran–Timmermann test against the right benchmark
- Explain data snooping and the deflated Sharpe ratio
- Compare Value-at-Risk forecasts by breach rate and by quantile score, and explain why these can disagree
- Name the specific risks of pretrained models: look-ahead through pretraining, survivorship, reproducibility
- Interpret the main results of Lectures 1–7 in the exam format

Abbreviations

AI artificial intelligence
AUC area under the ROC curve
BTC Bitcoin
DM Diebold–Mariano (test)
DSR deflated Sharpe ratio
EODHD EOD Historical Data
ES expected shortfall
ETF exchange-traded fund
EWMA-N exponentially weighted moving average, Normal quantile
GARCH-t generalised autoregressive conditional heteroskedasticity, Student-t errors
HAC heteroskedasticity- and autocorrelation-consistent
HAR heterogeneous autoregressive
HS historical simulation
LLM large language model
LR likelihood ratio (test)
ML machine learning
ROC receiver operating characteristic
MOM(k) k-day momentum rule
MA(s,l) moving-average crossover, s and l days
PSR probabilistic Sharpe ratio
PT Pesaran–Timmermann test
QGBM gradient-boosted quantile trees
QReg linear quantile regression
QS quantile score (pinball loss)
SHAP Shapley additive explanations
SR Sharpe ratio
VaR Value-at-Risk
XGBoost extreme gradient boosting

Reading and Tools

- Optimal shifts of a forecast and calibration diagnostics: Gneiting & Resin (2023)
- Quantile regression: Koenker & Bassett (1978); directional accuracy: Pesaran & Timmermann (1992)
- Data snooping: White (2000); Bailey & López de Prado (2014), *The deflated Sharpe ratio*
- Pretrained forecasting model: Ansari et al. (2025), *Chronos-2: from univariate to universal forecasting*; explanations: Lundberg & Lee (2017)
- Capital rules: Basel Committee, MAR32–33
- Video units on Quantinar: AI Meets Blockchain, Crypto-Backed P2P Lending
- Estimation: Python (`scikit-learn`, `xgboost`, `shap`, `arch`, `statsmodels`, `chronos-forecasting`); numbers marked "own" are estimated for this lecture

The Prediction Problem

- Target: tomorrow's direction $y_{t+1} = \mathbb{1}\{r_{t+1} > 0\}$; features x_t known at today's close: returns of the last three days, 5-, 20- and 60-day momentum, 5- and 20-day volatility, volume changes, gaps to the 50- and 200-day averages, day of the week (13 features)
- Three classifiers of increasing flexibility:
  - logistic regression (logit): $P(y_{t+1} = 1 \mid x_t) = \Lambda(\beta_0 + x_t'\beta)$, $\Lambda(u) = 1/(1 + e^{-u})$, with a ridge penalty
  - random forest: average of $B = 500$ trees, each grown on a bootstrap sample with random feature subsets
  - gradient boosting (XGBoost): additive trees, $F_m(x) = F_{m-1}(x) + \eta\, h_m(x)$, each h_m fitted to the gradient of the log-loss; $\eta = 0.05$, 300 trees of depth 3
- Walk-forward design: train on all data up to day τ, predict the next 63 days, move τ forward, refit; test 1 January 2020 – 22 October 2026 ($T = 2\,487$ days)
- Nothing from the test period ever enters a fit: this is the minimum standard for any claim of predictability

Measuring Predictive Skill

- Accuracy $= T^{-1}\sum_t \mathbb{1}\{\hat{y}_t = y_t\}$ must beat the base rate: Bitcoin rose on 50.6% of test days, so "always up" already scores 50.6%
- Brier score of the probabilities p_t: $\mathrm{BS} = T^{-1}\sum_t (p_t - y_t)^2$; the constant forecast $p_t = \bar{y}$ gives $\bar{y}(1-\bar{y}) \approx 0.250$
- Area under the receiver operating characteristic curve (AUC): probability that a random up-day gets a higher p_t than a random down-day; 0.5 = no skill
- Pesaran–Timmermann (PT) test: compares the hit rate $\hat{P}$ with the hit rate expected if forecasts and outcomes were independent, $P^* = \bar{y}\,\hat{y} + (1-\bar{y})(1-\hat{y})$:

$$\mathrm{PT} = \frac{\hat{P} - P^*}{\sqrt{\widehat{V}(\hat{P}) - \widehat{V}(P^*)}} \overset{a}{\sim} N(0,1)$$

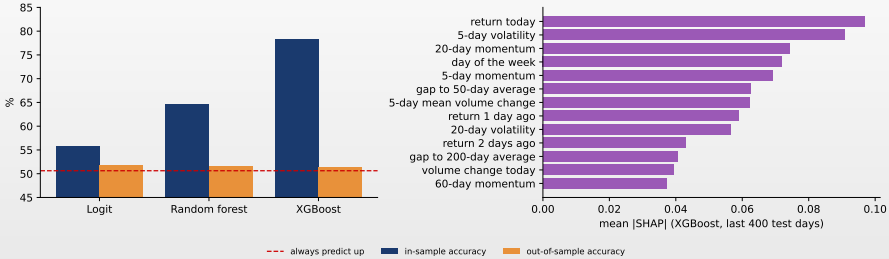
- Trading: long when $p_t > 0.5$, otherwise cash; 10 basis points (bp) cost per change of position; annualised Sharpe ratio (SR) $= \sqrt{365}\,\bar{g}/s_g$
- Sampling error of an annualised SR over Y years: $\mathrm{se} \approx \sqrt{(1 + \mathrm{SR}^2/2)/Y}$, about 0.4 for $Y = 6.8$

Results, 2020–2026 (own)

	accuracy (%)		AUC	Brier	PT	share	SR	max. draw-
	in-sample	out-of-sample			(p)	long		down (%)
Always up / buy and hold	–	50.6	0.500	0.250	–	100%	0.59	145
Logit	55.8	51.7	0.525	0.2515	0.03	85%	0.72	116
Random forest	64.6	51.6	0.514	0.2514	0.07	78%	0.81	161
XGBoost	**78.3**	51.3	0.508	0.2611	0.13	65%	0.63	147

- Out-of-sample accuracy is 0.7–1.1 percentage points above the base rate; only the logit passes the PT test at 5% (one-sided)
- Every Brier score is *worse* than the constant $p = 0.506$: the probabilities carry no usable information
- The more flexible the model, the larger the gap between in-sample and out-of-sample accuracy: XGBoost fits 78% of past days and 51% of new ones (overfitting)
- Sharpe ratios of 0.63–0.81 against 0.59: differences far below one standard error (≈ 0.4); the logit is long 85% of the time, so it is mostly buy-and-hold
- Accuracy by year falls from 55% (2020) to 46–51% (2025–2026): whatever was learnt does not persist

Overfitting and What the Model Uses (own)



- Shapley additive explanations (SHAP): the prediction is split into feature contributions, $f(x) = \phi_{\mathbf{0}} + \sum_j \phi_j(x)$, with ϕ_j the average marginal contribution of feature j over all orderings of the features
- The largest contributions come from today's return and 5-day volatility; the day of the week ranks fourth
- SHAP explains the *model*, not the market: a large $|\phi_j|$ for a model with no skill is not evidence that the feature predicts returns

Trading the Signals (own)



- The gains over buy-and-hold come in a few episodes (2020–2021, parts of 2022); in between, the curves move in parallel with buy-and-hold
- A strategy that is long 65–85% of the time inherits the market's drawdowns: maximum drawdowns of 116–161% in cumulative log return against 145% for buy-and-hold

Many Rules, One Winner

- If we test N rules with no skill, the best one still has a high Sharpe ratio. For N independent rules with true SR $= 0$, the expected maximum is approximately

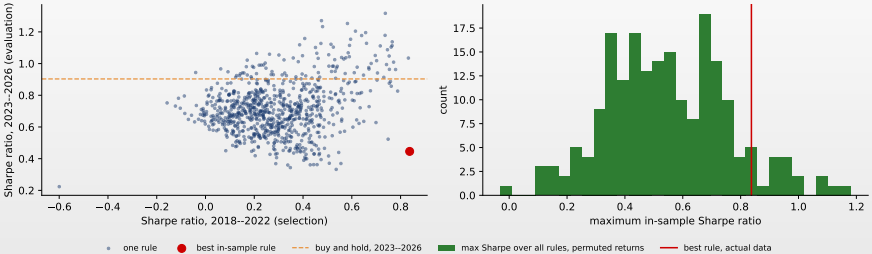
$$\mathrm{SR}_0 = \sqrt{V[\widehat{\mathrm{SR}}_n]}\left[(1-\gamma)\,\Phi^{-1}\!\left(1-\tfrac{1}{N}\right) + \gamma\,\Phi^{-1}\!\left(1-\tfrac{1}{Ne}\right)\right], \quad \gamma \approx 0.5772$$

- Probabilistic Sharpe ratio (PSR): probability that the true SR exceeds a benchmark SR^*, allowing for skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of daily returns over T days:

$$\mathrm{PSR}(\mathrm{SR}^*) = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \mathrm{SR}^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{\mathrm{SR}}^2}}\right)$$

- Deflated Sharpe ratio (DSR) $= \mathrm{PSR}(\mathrm{SR}_0)$: the benchmark is the best Sharpe ratio expected by chance (Bailey & López de Prado (2014))
- White's (2000) reality check: compare the best rule with the distribution of the best rule when no rule has skill
- Own design: 810 rules (560 moving-average crossovers MA(s,l) and 250 momentum rules MOM(k)), selected on 2018–2022, evaluated on 2023 – 23 October 2026; null distribution from 200 permutations of the selection-period returns

810 Trading Rules (own)



- Best rule in 2018–2022: MOM(2), SR $= 0.84$ against 0.04 for buy-and-hold. In 2023–2026: SR $= 0.45$ against 0.90, rank 779 of 810
- Under permuted returns the best of 810 rules reaches 0.55 on average and 0.93 at the 95th percentile: $p = 0.095$ for the winner
- PSR against zero $= 0.97$, but DSR $= 0.72$: once the search is counted, the winner is not significant
- Rank correlation of in-sample and out-of-sample SR: 0.15; only 12% of the rules beat buy-and-hold after 2022

Six Value-at-Risk Models

- One-day Value-at-Risk (VaR) at level α: the α-quantile q_t of r_t given information to $t-1$; $\alpha = 1\%$ and 5%; returns in %
- Classical (Lecture 3):
  - historical simulation (HS): empirical α-quantile of the last 500 returns
  - EWMA-N: $\sigma_t^2 = 0.94\,\sigma_{t-1}^2 + 0.06\,r_{t-1}^2$, $q_t = \Phi^{-1}(\alpha)\sigma_t$
  - GARCH(1,1)-t: $q_t = \mu + \sigma_t\, t_\nu^{-1}(\alpha)\sqrt{(\nu-2)/\nu}$, refitted every 21 days
- Machine learning:
  - linear quantile regression (QReg, Koenker & Bassett (1978)) on heterogeneous autoregressive (HAR) terms: $q_t = \beta_0 + \beta_1|r_{t-1}| + \beta_2 \mathrm{RV}_{t-1}^{(5)} + \beta_3 \mathrm{RV}_{t-1}^{(22)} + \beta_4 \mathrm{RV}_{t-1}^{(66)} + \beta_5 \min(r_{t-1}, 0)$, with $\mathrm{RV}^{(k)}$ the k-day root mean square return
  - gradient-boosted quantile trees (QGBM): 300 trees minimising the quantile loss on 9 features (returns, volatilities, volume change); rolling 1 500 days
- Pretrained: Chronos-2 (Ansari et al. (2025)), a transformer trained on a large corpus of real and synthetic series; it returns quantiles from 1% to 99% directly. *Zero-shot*: no parameter is fitted to Bitcoin; input = the last 512 daily returns
- Forecasts from 1 June 2022; test 1 January 2024 – 23 October 2026 ($T = 1\,027$ days)

How to Score a Quantile Forecast

- Breach indicator $h_t = \mathbb{1}\{r_t < q_t\}$; breach rate $\bar{h}$ should be close to α
- Kupiec likelihood ratio (LR) test of $\bar{h} = \alpha$; Christoffersen test of independent breaches (Lecture 3)
- Quantile score (QS), also called pinball loss:

$$\mathrm{QS}_\alpha(q, r) = (r - q)\,(\alpha - \mathbb{1}\{r < q\}) \geq 0$$

- The QS is *strictly consistent* for the quantile: its expectation is minimised only by the true conditional quantile. The breach rate is not: the forecast $q_t = -\infty$ on 99% of days and $+\infty$ on 1% has a perfect breach rate and is useless
- Diebold–Mariano (DM) test on the loss difference $d_t = \mathrm{QS}_t^A - \mathrm{QS}_t^B$:

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}_d / T}} \overset{a}{\sim} N(0,1), \qquad \hat{\Omega}_d = \text{HAC (Newey–West) long-run variance}$$

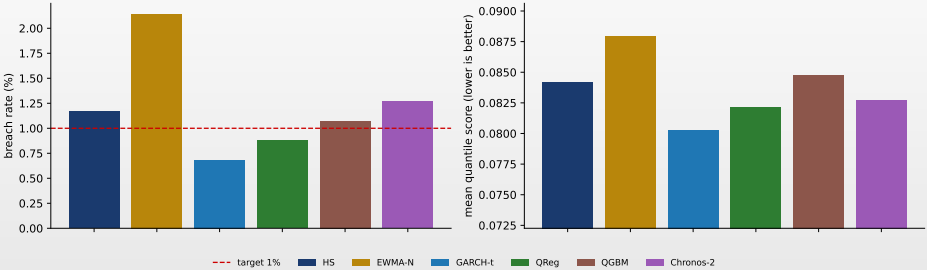
- Breach rate answers "is the level right on average?"; the QS answers "is the forecast close to the true quantile every day?"

Bitcoin VaR, 2024 – October 2026 (own)

	$\alpha = 1\%$ (10.3 breaches expected)					$\alpha = 5\%$			
model	breach %	Kupiec p	Christ. p	QS	DM p	breach %	Kupiec p	QS	DM p
HS	1.17	0.60	0.59	0.0842	0.45	5.36	0.61	0.2782	0.57
EWMA-N	**2.14**	**0.001**	0.33	0.0880	0.18	5.26	0.71	0.2785	0.18
GARCH-t	0.68	0.28	0.76	**0.0803**	–	5.65	0.35	**0.2748**	–
QReg	0.88	0.68	0.69	0.0822	0.71	3.89	0.09	0.2755	0.92
QGBM	1.07	0.82	0.63	0.0848	0.36	6.13	0.11	0.2875	0.07
Chronos-2	1.27	0.41	0.56	0.0827	0.54	5.16	0.81	0.2773	0.71

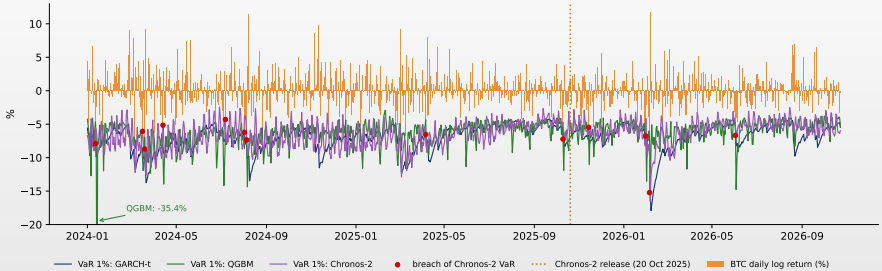
- DM p: two-sided test of equal mean QS against GARCH-t
- All models except EWMA-N pass the coverage tests. The best breach rate (QGBM, 1.07%) does not have the best QS; GARCH-t has the lowest QS with only 7 breaches out of 10.3 expected
- No model differs significantly from GARCH-t at 1%: a pretrained model with no Bitcoin-specific fitting is as good as, but not better than, a 40-year-old model
- The 5% level gives more breaches and more power; QGBM is the weakest there (p 0.07)

Breach Rate and Quantile Score Rank Models Differently (own)



- Closest to target: QGBM, HS, Chronos-2. Lowest QS: GARCH-t, QReg, Chronos-2
- A forecast can hit the right breach count while being too high on calm days and too low on volatile days; the QS penalises both

VaR Paths: Machine Learning Can Jump (own)



- On 14 January 2024 the tree model's VaR jumps from −7.3% to −35.4%; GARCH-t moves from −9.9% to −9.4%. The input: a −7.9% return after the spot Bitcoin exchange-traded fund (ETF) launch, then a 53% fall in volume, a combination the trees met only in crash periods
- Trees do not extrapolate smoothly: an unusual input lands in an extreme leaf. A risk manager needs bounds or a fallback rule

Pretrained Models and Look-Ahead

- A pretrained model may have seen the test period during training. Chronos-2 was released in October 2025; its training corpus may include Bitcoin prices for 2024–2025. Only forecasts after the release are clean
- Own check, 1% VaR, mean QS before and after 20 October 2025 (368 days after):

	HS	EWMA-N	GARCH-t	QReg	QGBM	Chronos-2
before release (659 days): QS	0.081	0.083	**0.077**	0.085	0.086	0.082
after release (368 days): QS	0.089	0.097	0.087	**0.077**	0.083	0.084
after release: breach %	0.82	2.17	0.54	1.09	1.09	1.09

- Chronos-2 ranks third in both periods: no sign of an advantage from leaked data, but 368 days and 4 expected breaches cannot prove its absence
- Large language models (LLMs) have the same problem in a stronger form: a model asked about 2024 "knows" what happened after 2024. Any back-test of an LLM must use dates after its training cut-off
- Other pretrained models: TimesFM 2.5 returns only deciles (10%–90%), so it cannot give a 1% VaR without extrapolation

Seven Limits

1. **Low signal**: daily return direction is close to a coin toss (base rate 50.6%, best accuracy 51.7%); flexibility adds variance, not signal
2. **Non-stationarity**: regimes change (2020–2021 boom, 2022 crash, 2024 ETF era); what is learnt in one regime may not transfer (accuracy by year, HS correction)
3. **Few tail observations**: a 1% quantile from 500 days rests on 5 points; tail models are noisy by construction (Lectures 3 and 5)
4. **Data snooping**: testing many models and reporting the best inflates performance; report the number of trials and deflate
5. **Look-ahead**: pretrained models and LLMs may have seen the test period; features must be known at the forecast time; selection must use only past data (BitMood, Lecture 6)
6. **Survivorship**: today's list of coins excludes the dead ones; results on survivors overstate performance
7. **Explanation $\neq$ cause**: SHAP and feature importances describe the model; causal claims need a design (event study, Lecture 6)

⊡ What ML does well here: flexible quantile forecasts that are competitive with GARCH, early-warning classifiers with many features (Lecture 4), text-based sentiment (Lecture 6), when evaluated out of sample with the right score

The Exam: Friday 27 November 2026

- Written, 90 minutes, closed book; one A4 formula sheet allowed; 60% of the final grade (seminar project with Prof. Alon: 40%)
- **Part A** (40% of the exam): 20 multiple-choice questions on concepts and on reading charts and model output
- **Part B** (60%): 3 short interpretation problems built on real output from the course (tables, charts, tests)
- No calculation is required. A complete answer in Part B has three parts:
  1. **What is measured**: name the statistic and its null hypothesis or benchmark
  2. **What the number says**: direction, size, significance, in the units of the problem
  3. **What it does not say**: the limitation (sample size, regime, multiple testing, look-ahead, association vs cause)
- Use the vocabulary of the course precisely: "significant at 5%", "out of sample", "fails the Kupiec test", "heavy tails"
- Preparation: the interpretation exercises at the end of each lecture and the practice exam with solutions

Review: Lectures 1–2

- **L1 Blockchains as data**
  - hash rate from difficulty, $\hat{H} = D \cdot 2^{32}/600$: a measure of security cost
  - supply rule: 21 million BTC, halving every 210 000 blocks; issuance inflation 0.82% after 2024
  - concentration: Herfindahl–Hirschman index (HHI) $= \sum s_i^2$; two issuers hold 85% of stablecoins
  - log-linear trends and their standard errors; data-quality checks before any analysis
- **L2 Stylised facts, efficiency, bubbles**
  - annualise with $\sqrt{365}$; heavy tails: Jarque–Bera, quantile–quantile plots
  - volatility clustering: Ljung–Box $Q(m)$ on r_t^2 vs on r_t
  - weak-form efficiency: variance ratio $\mathrm{VR}(k) = \mathrm{Var}(r_t^{(k)})/(k\,\mathrm{Var}(r_t))$, robust test
  - bubbles: log-periodic power law (LPPL), critical time t_c, fit is not forecast; Metcalfe's law $V \propto N^b$

Review: Lectures 3–4

- **L3 Volatility and tail risk**
  - GARCH(1,1): $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$; persistence $\alpha + \beta$, half-life; Student-t errors
  - VaR and ES; four methods (Normal, HS, EWMA, GARCH-t)
  - backtests: Kupiec (frequency), Christoffersen (independence), Basel traffic light
  - out of sample: what a short new period can and cannot show
- **L4 Stablecoins: depegs**
  - deviation in basis points $d_t = 10^4(P_t - 1)$; breach episodes; power-law tails
  - automated market makers (AMM): constant product $xy = k$ and StableSwap; pool composition as an early signal
  - event-level evaluation of alarms: detection rate, lead time, false alerts, utility

Review: Lectures 5–7

- **L5 The depeg pressure index (DPI)**
  - check loss $\rho_\tau(u) = u(\tau - \mathbb{1}\{u < 0\})$; 5% Quantile Lasso; penalty λ selected by generalised approximate cross-validation (GACV)
  - DPI = average selected penalty: a higher λ means stronger tail co-movement; under pressure fewer, stronger tail links remain
  - in-sample fit vs held-out skill
- **L6 Networks, sentiment and asset classes**
  - correlation distance $d_{ij} = \sqrt{2(1 - \rho_{ij})}$, minimum spanning tree, absorption ratio
  - Diebold–Yilmaz spillovers: total, TO, FROM, NET
  - event study: abnormal and cumulative abnormal returns (AR, CAR), cross-sectional tests
  - factor analysis and classification: are cryptos an asset class?
  - sentiment indices and out-of-sample selection
- **L7 Machine learning, AI and limits**
  - walk-forward evaluation; base rate, Brier, AUC, PT test; in-sample vs out-of-sample
  - data snooping and the deflated Sharpe ratio
  - breach rate vs quantile score; look-ahead in pretrained models

Worked Example: a Model Answer

- **Question**: "Bitcoin 1% VaR, 2024 – October 2026: QGBM has a breach rate of 1.07%, GARCH-t of 0.68%; GARCH-t has the lower mean QS (DM p = 0.36). Which model is better?"
- **What is measured**: the breach rate checks the average level of the 1% VaR; the QS measures daily closeness to the true quantile; the DM test asks whether the mean QS difference is zero
- **What the numbers say**: QGBM is closer to 1% on average; GARCH-t is slightly conservative but tracks the quantile better day by day
- **Why they disagree**: a model can hit 1% on average by being too high on calm days and too low on volatile days; only the QS penalises that
- **What it does not say**: that GARCH-t is better. With DM p = 0.36 the difference is not significant, and 1 027 days hold only about 10 expected breaches

Summary

- Direction of Bitcoin: ML accuracy 51.3–51.7% against a base rate of 50.6%; Brier scores worse than a constant; XGBoost 78% in sample, 51% out of sample
- Trading rules: the best of 810 rules in 2018–2022 ranks 779th afterwards; DSR 0.72 once the search is counted
- VaR: classical, ML and pretrained models are statistically indistinguishable by QS at 1%; GARCH-t has the lowest QS, QGBM the best breach rate
- Chronos-2 forecasts without fitting and keeps its rank after its release date; look-ahead must still be ruled out by design

Interpretation Exercises

1. A random forest classifies tomorrow's Bitcoin direction with 64.6% accuracy in sample and 51.6% out of sample; 50.6% of days are up days. Which number would you report, and against which benchmark?
2. A model's Brier score is 0.2611 and that of the constant forecast 0.2500. What does this say about the model's probabilities, even if its accuracy exceeds 50%?
3. A consultant shows a momentum rule with Sharpe ratio 0.84 over five years. You learn it is the best of 810 rules tried. Explain the deflated Sharpe ratio of 0.72 to a client.
4. QGBM has a 1% breach rate of 1.07%, GARCH-t of 0.68%; the QS is lower for GARCH-t. Which model is better? Why can the two criteria disagree?
5. A pretrained model released in October 2025 is back-tested on 2024–2025 and beats GARCH. What is your main concern, and how would you address it?

⊡ Good luck in the exam on 27 November!

Appendix: Further Material

Optional material for self-study. It is **not examined**.

- Correcting a Forecast with Its Own Errors
- Bias against Estimation Cost
- Monte Carlo: Raw, Half, Full (own)
- Recalibrating the Bitcoin Forecasts (own)
- Breach Rate and Loss after Correction (own)
- Recalibration and the Basel Traffic Light (own)

Correcting a Forecast with Its Own Errors

- Forecast errors on a calibration window of n days: $e_t = r_t - q_t$. If the forecast were right, $P(e_t < 0) = \alpha$
- Fitted correction: $\hat{c} = \hat{Q}_\alpha(e_{t-n}, \ldots, e_{t-1})$, the empirical α-quantile of the errors; then $P(e < \hat{c}) \approx \alpha$ on the calibration window
- Corrected forecast with intensity λ:

$$q_t^{(\lambda)} = q_t + \lambda \hat{c}, \qquad \lambda = 0 \text{ (Raw)}, \ \tfrac{1}{2} \text{ (Half)}, \ 1 \text{ (Full)}$$

- Static: $\hat{c}$ fitted once on the 500 errors before the test and kept fixed. Rolling: refitted every day on the last 500 errors
- Full correction is what a conformal or "backtest-and-adjust" procedure does: it makes the calibration breach rate exactly α
- Question: does it lower the QS on *future* days? A breach rate closer to α is not the same thing

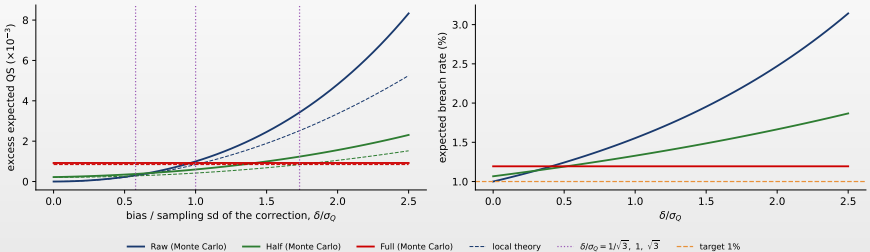
Bias against Estimation Cost

- Let δ be the true shift that would minimise the expected QS (the bias of the raw forecast) and $\hat{c} = \delta + \varepsilon$, $V(\varepsilon) = \sigma_Q^2$. With f the density at the quantile, a second-order expansion gives the excess expected loss

$$\Delta(\lambda) \approx \frac{f}{2}\left[(1-\lambda)^2\delta^2 + \lambda^2\sigma_Q^2\right]$$

- Estimation cost under dependence: $\sigma_Q^2 \approx \dfrac{\Omega_h}{nf^2}$, $\Omega_h = \sum_{k=-\infty}^{\infty} \mathrm{Cov}(h_t, h_{t+k})$, the long-run variance of the breach indicator; it falls as $1/n$
- Optimal intensity: $\lambda^* = \dfrac{\delta^2}{\delta^2 + \sigma_Q^2}$ (shrinkage, as in a James–Stein estimator)
- Comparisons:
  - Full beats Raw $\iff \delta^2 > \sigma_Q^2$
  - Half beats Full $\iff \delta^2 < 3\sigma_Q^2$; Half beats Raw $\iff \delta^2 > \sigma_Q^2/3$
  - Half is optimal when $\delta^2 = \sigma_Q^2$
- δ refers to the test period. If the bias changes between calibration and test, the fitted $\hat{c}$ transfers badly, whatever n is

Monte Carlo: Raw, Half, Full (own)



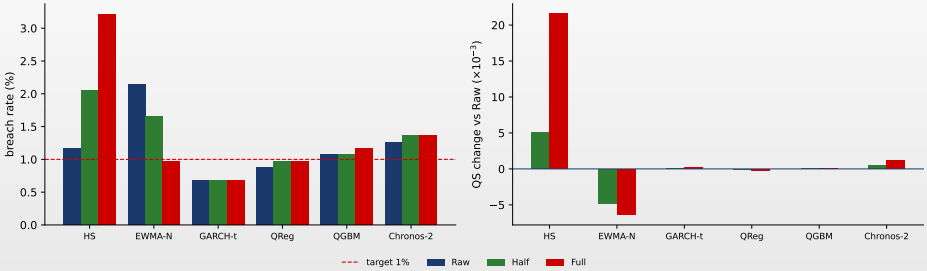
- Design: returns i.i.d. Student-t_5, $\alpha = 1\%$, $n = 500$, raw forecast shifted by δ, 20 000 replications; dashed lines = local theory
- Simulated crossings: Half beats Raw from $\delta/\sigma_Q = 0.65$ (theory 0.58), Half beats Full up to 1.40 (theory 1.73), Full beats Raw from 0.95 (theory 1)
- With no bias, Full still raises the expected QS and gives 1.19% breaches: the 1% empirical quantile of 500 errors rests on 5 observations

Recalibrating the Bitcoin Forecasts (own)

$\alpha = 1\%$	$\hat{c}$	$\hat{c}^2/\widehat{\mathrm{se}}^2$	breach % (static)			QS (static)		
			Raw	Half	Full	Raw	Half	Full
HS	1.91	2.18	1.17	2.04	**3.21**	0.0842	0.0893	**0.1058****
EWMA-N	−1.14	0.36	2.14	1.66	0.97	0.0880	0.0831*	0.0816
GARCH-t	−0.06	0.00	0.68	0.68	0.68	0.0803	0.0804	0.0805
QReg	0.63	0.57	0.88	0.97	0.97	0.0822	0.0821	0.0820
QGBM	0.08	0.01	1.07	1.07	1.17	0.0848	0.0848	0.0849
Chronos-2	0.33	0.05	1.27	1.36	1.36	0.0827	0.0833	0.0839

- $\widehat{\mathrm{se}}$: block bootstrap (blocks of 20 days). Stars: DM test against Raw, * 5%, ** 1%
- HS: in mid-2022 – 2023 its 500-day window still held the 2022 crash, so it was too conservative and $\hat{c} > 0$. The correction raised VaR just before the more volatile 2024: breaches rise to 3.2% and the QS gets 26% worse. At 5% the same correction gives 14.0% breaches
- EWMA-N: the Normal quantile is too thin; $\hat{c} < 0$ transfers, and both corrections help
- Models with $\hat{c}^2/\widehat{\mathrm{se}}^2$ near zero (GARCH-t, QGBM, Chronos-2): the correction is noise and changes almost nothing
- Rolling daily correction limits the HS damage: QS 0.0908 instead of 0.1058

Breach Rate and Loss after Correction (own)



- Left: correction moves breach rates in both directions. Right: QS change against Raw ($\times 10^{-3}$); positive = worse
- Same tool, opposite outcomes: the sign depends on whether the calibration-period bias persists into the test period

Recalibration and the Basel Traffic Light (own)

- Basel traffic light for a 1% VaR: a zone is yellow once the binomial probability of the breach count reaches 95%, red at 99.99% (Basel Committee, MAR32–33). For 250 days: green 0–4, yellow 5–9, red 10+. For a crypto year of 365 days: green 0–6, yellow 7–12, red 13+
- A lower (more negative) VaR means more capital: the capital requirement scales with the VaR or ES level times a zone multiplier (higher in yellow and red)

	breaches 2024 (366 days)			breaches 2025		mean VaR 2024–2025 (%)	
	Raw	Half	Full	Raw	Full	Raw	Full
HS	6	**10**	**17**	3	**11**	−6.3	−4.4
EWMA-N	**7**	6	3	**10**	3	−5.7	−6.8
GARCH-t	2	2	2	3	3	−7.1	−7.1
QReg	3	4	4	2	2	−7.6	−7.0
QGBM	3	3	4	4	4	−6.5	−6.4
Chronos-2	**7**	**8**	**8**	3	3	−6.7	−6.4

- Bold = yellow (7–12) or red (13+). EWMA-N: Full moves yellow to green in both years, but its VaR is 1.1 points deeper, so more capital
- HS: Full moves green to red in 2024 (17 breaches) with a *shallower* VaR: less capital, worse model
- Zone, capital and forecast quality are three different outcomes of the same correction; report all three